

Software Requirements Specification (SRS)

Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains

Document Identifier: SRS-IR-ETA-2026-V1.0

Target Organization: Indian Railways / Centre for Railway Information Systems (CRIS)

System Title: Dynamic Train Arrival Prediction Engine (DTAPE)

Status: Approved for Architectural Baseline

1. Introduction

1.1 Purpose

This document provides a comprehensive Software Requirements Specification (SRS) for the **Dynamic Train Arrival Prediction Engine (DTAPE)**. The system transitions train arrival forecasting from static timetable extrapolation to an intelligent, event-driven, spatio-temporal machine learning architecture. It serves operational control offices, train dispatchers, station management, rolling stock depots, and public enquiry platforms across all 17 operational zones of Indian Railways.

1.2 Document Conventions

- **Requirement Identifiers:** System requirements follow the taxonomy [REQ-TYPE-MODULE-ID]:
 - FR: Functional Requirement
 - NFR: Non-Functional Requirement
 - IFR: Interface Requirement
 - DR: Data Requirement
- **Priority Levels:** Prioritized according to the MoSCoW framework:
 - **MUST:** Mandatory baseline functionality for minimum viable production deployment.
 - **SHOULD:** High-priority capability required for complete operational efficiency.
 - **COULD:** Desirable capability to be phased in post-trunk line rollout.
 - **WON'T:** Explicitly excluded from the current architectural scope.

1.3 Intended Audience and Reading Suggestions

- **System Architects and ML Engineers:** Focus on Section 3 (System Features & Functional Requirements), Section 5 (Algorithmic & Machine Learning Specifications), and Section 6 (Data & Schema Specifications).
- **Enterprise Integration Teams (CRIS/COA/NTES):** Focus on Section 4 (External Interface Requirements) and Section 6.3 (API Protocols).
- **Operational Controllers and Station Staff:** Focus on Section 2 (Overall Description) and Section 3.6 (Web Operational Dashboard).
- **DevOps and Site Reliability Engineers:** Focus on Section 5 (Non-Functional

Requirements) and Section 7 (Scalability & Infrastructure).

1.4 Project Scope

DTAPE is a real-time predictive backend engine coupled with operational web consoles and public-facing distribution APIs. The system:

- Ingests real-time satellite telemetry, signaling status, speed restrictions, and dispatch logs.
- Evaluates dynamic network graph interactions, train precedence hierarchies, and weather impacts.
- Forecasts point-in-time ETAs and calibrated probabilistic arrival intervals for every coaching train at all downstream stations.
- Distributes predictions with low latency to the Control Office Application (COA), National Train Enquiry System (NTES), Crew Management System (CMS), and Passenger Information Display Systems (PIDS).

Out of Scope:

- Direct automated dispatching or computer-controlled throwing of physical track switches.
- Passenger ticketing and reservation transactions (handled separately by PRS/IRCTC).
- Locomotive automated driving or direct throttle actuation.

1.5 Definitions, Acronyms, and Abbreviations

Acronym	Expansion	Description
ABS	Automatic Block Signaling	Continuous track signaling dividing sections into automated braking blocks.
CLS	Central Location Server	Centralized CRIS gateway ingesting all locomotive telematics.
CMS	Crew Management System	Enterprise platform managing loco pilot and guard duty rosters.
COA	Control Office Application	Digital charting platform used by divisional train dispatchers.
CQR	Conformalized Quantile	Statistical framework guaranteeing valid

	Regression	distribution-free prediction bounds.
CRIS	Centre for Railway Information Systems	IT and software development enterprise for Indian Railways.
EA	Engineering Allowance	Timetable buffer absorbing temporary speed restrictions and civil maintenance.
EI	Electronic Interlocking	Microprocessor-based track and switch logic ensuring route safety.
ETA	Expected Time of Arrival	Dynamically forecasted timestamp of train arrival at a specified node.
GAGAN	GPS-Aided GEO Augmented Navigation	Indian satellite-based augmentation system providing meter-level accuracy.
ICMS	Integrated Coaching Management System	Tracks rake composition, maintenance schedules, and coach punctuality.
MPS	Maximum Permissible Speed	Legally authorized maximum velocity permitted across an engineered section.
MSRT	Minimum Sectional Running Time	Transit time achieved running continuously at MPS without deceleration.
NSRT	Normal Sectional Running Time	Timetabled transit time operating at booked speed.
NTES	National Train Enquiry	Public traveler information

	System	platform for Indian Railways.
OHE	Overhead Equipment	High-voltage electric traction wires suspended above the railway tracks.
PIDS	Passenger Information Display System	Electronic audio-visual display boards on station platforms.
RSTGCN	Railway Spatio-Temporal Graph CNN	Deep learning graph architecture capturing network-wide delay diffusion.
RTIS	Real-Time Train Information System	Satellite-based cab device transmitting periodic telemetry to ISRO/CRIS.
TRT	Traffic Recovery Time	Schedule buffer placed upstream of major terminal junctions to recover delays.
TSR	Temporary Speed Restriction	Caution orders mandating localized speed reductions due to track works.
WTT	Working Time Table	Official internal operating timetable specifying operational constraints.

2. Overall Description

2.1 Product Perspective and Context

DTAPE operates within the distributed enterprise software ecosystem of Indian Railways. It functions as a computational intelligence layer between real-time telemetry providers, existing administrative databases, and consumption channels.

Upstream Data Provider	Communication Channel / Protocol	Transmitted Information
------------------------	----------------------------------	-------------------------

RTIS Locomotive Units	ISRO S-Band Satellite / 4G LTE to CLS	High-frequency GPS/GAGAN coordinates, speed, heading, and geofence events.
Control Office Application	Apache Kafka via CDC / gRPC	Dispatcher decisions, loop-line diversions, planned overtakes, and platform berths.
Caution Order System (COIMS)	RESTful HTTPS JSON Push	Active Temporary Speed Restrictions (TSRs) with kilometeric bounds and speed caps.
Station Interlocking / Data Loggers	MQTT / Modbus over TCP	Real-time signal aspect states, block occupancies, and track route lockings.
Environmental & Fog Feeds	HTTPS REST API	Fog-PASS activation flags, station visibility indices, and extreme precipitation.
WTT Master Repositories	PostgreSQL Batch Replicas	Static timetables, booked speeds, MSRT, NSRT, EA, and TRT margins.

Downstream Consumer	Delivery Protocol	Delivered Information
COA Dispatcher Workstations	Low-latency gRPC Streams	60-minute lookahead conflict projections, trajectory paths, and delay recovery bounds.
NTES Passenger Mobile App	HTTPS REST / CDN Cache	Dynamic station arrival countdowns, platform ETAs, and delay status.
Station Platform Displays (PIDS)	WebSockets / MQTT Edge Broker	Live station arrival countdowns and platform occupancy tracking.

Crew Management System (CMS)	RESTful Event Webhooks	Crew duty-hour violation warnings and projected mid-section relief risks.
------------------------------	------------------------	---

2.2 Product Functions

- Continuous Telematics Ingestion and Fusion:** Ingesting, deduplicating, and validating telemetry packets across 10,000 active rakes simultaneously.
- Dynamic Network Graph Representation:** Maintaining an in-memory spatio-temporal directed multi-graph of track topology, signal blocks, and active line capacities.
- Multi-Scale Delay Diffusion Modeling:** Processing macro-scale congestion propagation using Graph Neural Networks and micro-scale running dynamics using Quantile Regression.
- Domain Rule and Operational Heuristic Execution:** Dynamically recalculating engineering allowances, precedence overtakes, and weather speed caps based on physical ground conditions.
- Calibrated Probabilistic Forecasting:** Generating arrival time confidence intervals ($\tau \in [0.10, 0.90]$) using Conformalized Quantile Regression.
- High-Throughput, Low-Latency Serving:** Distributing dynamic ETAs with sub-200 millisecond API responses across national passenger enquiry and control room endpoints.

2.3 User Classes and Operating Characteristics

User Persona	Technical Competency	Operational Role in System	Key Requirements
Section Train Controller	High (Railway Operations)	Manages train routing, overtakes, and priority dispatching on divisional sections.	30-to-120 minute lookahead curves, platform conflict warnings, and loop-line dwell projections.
Station Master / Yard Master	Medium (Station Management)	Prepares platform berths, manages passenger flow, and dispatches shunting.	Station approach arrival countdowns, platform throat clearance alerts, and berthing ETAs.
Crew Controller (CMS Staff)	Medium (Administration)	Monitors loco pilot and train manager	Automated alerts for trains projected

		statutory duty durations (10-hour rule).	to exceed crew duty thresholds before reaching relief points.
Passenger / End User	Low to Medium	Accesses public apps for journey planning and connection management.	Clear point-in-time ETA, delay duration, and bounded confidence intervals (e.g., \$pm 5\$ min).
System Administrator / MLOps	High (Software & Data Systems)	Maintains infrastructure health, tracks model drift, and manages canary deployments.	System health metrics, feature drift metrics (PSI), inference latency dashboards, and fallback toggles.

2.4 Design and Implementation Constraints

- **Geographical Constraints:** System must accommodate 69,000 route kilometers, 4,735 passenger-handling stations, and diverse topographies (plains, ghat sections, coastal zones, desert routes).
- **Network Connectivity Gaps:** Satellite fallback mechanisms must operate seamlessly across areas with intermittent 4G/LTE coverage.
- **Legacy System Formats:** Ingestion layers must interface with legacy data stores (relational databases, structured text logs, legacy serial feeds) without interrupting existing operational flows.
- **Real-Time SLAs:** Predictive outputs must be calculated and served within strict latency envelopes to maintain operational relevance for moving locomotives.

2.5 Assumptions and Dependencies

- The CRIS Central Location Server (CLS) maintains continuous, authenticated streaming access to RTIS cab telemetry.
- Locomotives are mapped to specific scheduled train numbers via the Locomotive Link and Distribution operational system.
- Static timetable updates (new WTT publications) are made available programmatically prior to formal operational enforcement.

3. System Features and Functional Requirements

3.1 Module 1: High-Throughput Telematics & Ingestion Engine

3.1.1 Description

Captures, authenticates, decodes, and standardizes multi-source streaming telemetry from locomotives, control offices, and wayside signaling infrastructure.

3.1.2 Functional Requirements

Requirement ID	Description	Priority	Acceptance Criteria
FR-ING-01	The system MUST ingest high-frequency RTIS positional packets via an enterprise message broker (Kafka).	MUST	Ingestion latency ≤ 500 ms for a continuous stream of 10,000 active locomotives.
FR-ING-02	The system MUST ingest automated geofence event triggers representing station arrivals, departures, and run-through passings.	MUST	Event records processed and state-transitioned within 1.0 second of generation.
FR-ING-03	The system MUST ingest live Caution Orders (e-TSR) published by the Caution Order Information Management System (COIMS).	MUST	Dynamic speed restrictions mapped to spatial track segments within 30 seconds of issuance.
FR-ING-04	The system MUST ingest Fog-PASS device activations and station Visibility Test Object (VTO) declarations.	MUST	Track sections tagged with low-visibility operational constraints within 60 seconds of

			broadcast.
FR-ING-05	The system MUST validate incoming data payloads against corrupted coordinates, out-of-order timestamps, and telemetry drops.	MUST	Invalid packets discarded or routed to dead-letter queues with $< 0.01\%$ false rejection of valid data.

3.2 Module 2: Spatio-Temporal Feature Engineering Pipeline

3.2.1 Description

Transforms raw positional and tabular data streams into structured feature vectors and dynamic graph embeddings that capture physical running states and network-level interactions.

3.2.2 Functional Requirements

Requirement ID	Description	Priority	Acceptance Criteria
FR-FEA-01	The system MUST execute map-matching of geodetic coordinates to track centerline vectors (GeoJSON/Shapefiles).	MUST	Match locomotive coordinates to specific track alignments with $\geq 99.5\%$ spatial accuracy.
FR-FEA-02	The system MUST compute rolling kinematic features: instantaneous velocity, 3-minute sectional acceleration, and halt durations.	MUST	Metrics recalculated on every incoming telemetry packet within an Apache Flink stream.

FR-FEA-03	The system MUST compute dynamic spatial and temporal headways between consecutive trains operating on shared track circuits.	MUST	Headway separation updated within 1 second of positional state change.
FR-FEA-04	The system MUST maintain a low-latency feature cache storing current running states and rolling aggregates.	MUST	Key-value feature vector retrieval latency ≤ 5 ms for inference requests.

3.3 Module 3: Dynamic Multi-Scale ETA Prediction Engine

3.3.1 Description

Evaluates network-wide congestion diffusion and train-level running dynamics using a two-tier machine learning architecture to forecast arrival times and uncertainty bounds.

3.3.2 Functional Requirements

Requirement ID	Description	Priority	Acceptance Criteria
FR-MOD-01	The system MUST execute a macro-scale Railway Spatio-Temporal Graph Convolutional Network (RSTGCN) across active network corridors.	MUST	Generates updated station node congestion embeddings every 60 seconds across 4,735 stations.
FR-MOD-02	The system MUST compute micro-scale train	MUST	Predicts section transit times across all downstream

	travel times using an ensemble of Gradient Boosted Decision Trees and Temporal Convolutional Networks.		stations for active trains in < 50 ms.
FR-MOD-03	The system MUST output point-in-time median arrival forecasts alongside 10th and 90th percentile quantile predictions.	MUST	Pinball loss minimization implemented natively within the training and inference pipeline.
FR-MOD-04	The system MUST calibrate quantile outputs using Mondrian Conformalized Quantile Regression (CQR).	MUST	Achieves empirical coverage of $\geq 80\%$ on holdout test datasets across all service classes.
FR-MOD-05	The system MUST incorporate fallback heuristic logic if ML inference fails or times out.	MUST	Seamlessly falls back to physics-based kinematic running time calculation within 10 ms.

3.4 Module 4: Operational Rules & Railway Logic Integration

3.4.1 Description

Embeds statutory Indian Railways operational constraints, timetable engineering rules, and controller precedence protocols directly into the prediction pipeline.

3.4.2 Functional Requirements

Requirement ID	Description	Priority	Acceptance Criteria
----------------	-------------	----------	---------------------

FR-RUL-01	The system MUST dynamically adjust Engineering Allowance (EA) recovery based on real-time TSR records.	MUST	If zero TSRs exist in a section, EA is credited toward delay recovery; if active TSRs exist, speed penalties are applied.
FR-RUL-02	The system MUST adjust Traffic Recovery Time (TRT) as trains approach major junction approach sectors.	MUST	Scheduled TRT buffer deducted from cumulative delay only when entering designated terminal zones.
FR-RUL-03	The system MUST apply precedence overtaking penalties based on train hierarchy matrices.	MUST	Applies turnout deceleration (15/30 km/h), dwell, and acceleration penalties when an overtake is projected.
FR-RUL-04	The system MUST enforce adverse weather speed caps (60–75 km/h) upon receipt of Fog-PASS alerts or low-visibility triggers.	MUST	Sectional travel times recalculated using weather-constrained velocity profiles across affected zones.

3.5 Module 5: API Gateway and External Distribution Services

3.5.1 Description

Exposes secure, high-throughput, horizontally scalable interfaces to serve predictions to internal operational systems and public consumer channels.

3.5.2 Functional Requirements

Requirement ID	Description	Priority	Acceptance Criteria
----------------	-------------	----------	---------------------

FR-API-01	The system MUST expose a high-performance gRPC endpoint for divisional COA electronic control chart integration.	MUST	Delivers full downstream trajectory updates within 50 ms per train query.
FR-API-02	The system MUST provide RESTful JSON endpoints querying train ETAs by Train Number, Station Code, and Journey Date.	MUST	P99 response latency ≤ 200 ms under a baseline load of 20,000 requests per second.
FR-API-03	The system MUST expose a WebSocket streaming channel for real-time station platform countdown displays (PIDS).	MUST	Pushes arrival countdown changes to connected station platform clients within 1.0 second of recalculation.
FR-API-04	The system MUST implement token-based authentication (OAuth2 / JWT) and rate limiting across all public-facing interfaces.	MUST	Rejects unauthorized requests with HTTP 401; throttles requests exceeding tenant quotas with HTTP 429.

3.6 Module 6: Web-Based Operational Control Room Dashboard

3.6.1 Description

Provides an interactive, real-time visualization platform for section controllers, station masters, and divisional operational heads.

3.6.2 Functional Requirements

Requirement ID	Description	Priority	Acceptance Criteria
FR-UI-01	The dashboard MUST render dynamic train trajectories overlaid on a continuous time-distance string chart (digital control chart).	MUST	Trajectory curves update synchronously with incoming RTIS telemetry.
FR-UI-02	The dashboard MUST highlight projected platform conflicts and yard-throat deadlocks with visual alerts.	MUST	Alerts displayed ≥ 30 minutes in advance of projected arrival conflicts.
FR-UI-03	The dashboard MUST display operational confidence intervals around each train's predicted arrival curve.	SHOULD	Renders shaded bounds representing the 10th-to-90th percentile arrival windows.
FR-UI-04	The dashboard MUST support interactive "what-if" dispatch scenario modeling (e.g., simulating the impact of loop-line overtakes).	COULD	Computes and displays network-wide delay impact for controller scenarios within 2.0 seconds.

4. External Interface Requirements

4.1 User Interfaces

- **Operational Control Console (Web):** Single-Page Application (SPA) optimized for dual-monitor control room setups (resolution ≥ 1920 × 1080). Dark-mode

interface designed to minimize visual fatigue, rendering interactive vector-based track topologies, digital control charts, and color-coded alert drawers.

- **Station Master / Terminal Console:** Responsive tablet- and desktop-friendly view focusing on platform berthing occupancy, 60-minute station approach countdowns, and turn-around rake status.

4.2 Hardware Interfaces

- **RTIS Cab Unit Transceivers:** The platform interfaces with locomotive cab equipment via standardized protocol envelopes transmitted over ISRO S-band MSS links and 4G/LTE modems. The system accepts standard NMEA-0183 / GAGAN binary sentences converted into structured JSON/Protobuf packets at the ground gateway.

4.3 Software Interfaces

- **Centre for Railway Information Systems (CRIS) Central Location Server (CLS):** Direct TCP/Kafka connection consuming pre-processed locomotive telemetry.
- **Control Office Application (COA):** Direct bi-directional integration with divisional Oracle/PostgreSQL databases via Kafka Change Data Capture (CDC) connectors to mirror dispatch decisions and route lockings.
- **National Train Enquiry System (NTES):** Outbound REST push/pull API providing cached ETA values for public passenger query servicing.
- **Geospatial Infrastructure:** Ingestion of open railway topology schemas (DataMeet and OpenRailwayMap GeoJSON) for topological network graphs.

4.4 Communications Interfaces

- **Internal Microservice Mesh:** High-performance RPC (gRPC) with Protocol Buffers over HTTP/2, managed within a Kubernetes cluster.
- **External Client Interfaces:** HTTPS (TLS 1.3) REST endpoints for mobile applications; persistent WebSockets (WSS) for platform displays and web consoles.

5. Non-Functional Requirements (NFRs)

5.1 Performance Requirements

Metric	Target Specification	Measurement Context
End-to-End Latency	≤ 2.0 seconds	Time elapsed from locomotive GAGAN coordinate generation to database availability.
Model Inference Latency	≤ 50 milliseconds	Micro-scale Tier 2 batch inference per train across 10 upcoming stations.

Graph Refresh Latency	≤ 60 seconds	Macro-scale Tier 1 RSTGCN graph embedding refresh across all 4,735 stations.
API Response Time	≤ 150 ms (P95), ≤ 200 ms (P99)	Read throughput under peak load (20,000 req/s).
Stream Processing Throughput	$\geq 50,000$ events/second	Peak processing capability across the Apache Flink ingestion pipeline.

5.2 Scalability and Capacity Requirements

- **Active Train Concurrency:** The system MUST continuously track and forecast ETAs for up to 10,000 active coaching and freight trains simultaneously without performance degradation.
- **Station Node Capacity:** The system MUST maintain spatial representations for 4,735 passenger stations and up to 15,000 intermediate signal block divisions.
- **Horizontal Scalability:** Microservice pods and model serving instances MUST auto-scale horizontally based on CPU utilization ($\geq 70\%$) and Kafka consumer lag metrics.

5.3 Reliability, Availability, and Fault Tolerance

- **System Availability:** 99.99% operational uptime across primary and secondary disaster recovery regions.
- **Failover Time:** Automated recovery with a Recovery Time Objective (RTO) ≤ 30 seconds and a Recovery Point Objective (RPO) ≤ 5 seconds.
- **Graceful Degradation:** If the GPU-accelerated RSTGCN inference engine fails, the system MUST fallback automatically to tabular gradient boosting models, and subsequently to WTT kinematic calculations, ensuring unbroken service availability.

5.4 Security and Compliance Requirements

- **Encryption Standards:** All data in transit MUST use TLS 1.3 encryption. All persistent data at rest MUST be encrypted using AES-256.
- **Authentication and Authorization:** Role-Based Access Control (RBAC) enforced via OpenID Connect (OIDC) and short-lived JWT tokens. Operational dispatch endpoints restricted to authenticated Indian Railways intranet IP ranges.
- **Compliance:** Architecture must adhere to the Indian Government Cyber Security Guidelines issued by CERT-In and Indian Railways IT Security Policies.

5.5 MLOps, Drift Detection, and Maintainability

Feature	Specification	Implementation Strategy
Data Drift Detection	Population Stability Index	Continuous monitoring of

	(PSI) > 0.2 triggers alerts.	input feature distributions (speed, delay, headways).
Concept Drift Monitoring	Rolling MAE degradation $\geq 15\%$ triggers retraining.	Comparison of real-time predictions against realized station arrival timestamps.
Retraining Schedule	Weekly for tabular regressors; Monthly for graph networks.	Automated training pipelines executed on historical operational data lakes.
Deployment Strategy	72-hour shadow evaluation followed by canary rollout.	Production candidate must demonstrate $\geq 5\%$ MAE improvement and valid conformal coverage.

6. Algorithmic and Mathematical Formulation

6.1 Network Graph Construction

The national railway network is modeled as a directed multi-graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{W}_s, \mathcal{W}_t)$, where $\mathcal{V}$ defines stations and block signals, $\mathcal{E}$ defines interconnecting track blocks, $\mathcal{W}_s$ represents static physical parameters (track length, curvature, gradient, MPS), and $\mathcal{W}_t$ tracks dynamic states (signal aspects, active caution orders, headways).

6.2 Delay Diffusion Formulation

The downstream delay δ_j^k of train k arriving at station v_j from station v_i is modeled as:

$$\delta_j^k = \max\left(0, \delta_i^k + \Delta t_{\text{actual}}^k(i, j) - \Delta t_{\text{sched}}^k(i, j) - \beta_{ij}^k\right) + \sum_{m \in \mathcal{P}_k(i, j)} \gamma_{m \rightarrow k} (h_{\min} - h_{mk})^+$$

where:

- δ_i^k is the departure delay of train k at station v_i .
- $\Delta t_{\text{actual}}^k(i, j)$ and $\Delta t_{\text{sched}}^k(i, j)$ represent the actual and scheduled section transit times.
- β_{ij}^k is the dynamic recovery buffer (Engineering Allowance and Traffic Recovery Time).
- $\mathcal{P}_k(i, j)$ denotes the set of preceding or conflicting trains sharing the route.
- h_{mk} is the operational headway separation between train m and train k , and $h_{\min}$ is the signaling headway threshold.
- $\gamma_{m \rightarrow k} \in [0, 1]$ represents the priority penalty coefficient reflecting train

class hierarchy.

6.3 Uncertainty Quantification via Conformalized Quantile Regression (CQR)

To ensure reliable, distribution-free prediction bounds, the model trains quantile regressors $\hat{q}_{\tau}(X)$ targeting quantiles $\tau \in \{\alpha/2, 1 - \alpha/2\}$ using the pinball loss function:

$$\mathcal{L}_{\tau}(y, \hat{q}_{\tau}) = \sum_{i: y_i \geq \hat{q}_{\tau}(X_i)} \tau |y_i - \hat{q}_{\tau}(X_i)| + \sum_{i: y_i < \hat{q}_{\tau}(X_i)} (1 - \tau) |y_i - \hat{q}_{\tau}(X_i)|$$

Using a calibration set $(X_i, Y_i)_{i=1}^{n_{\text{cal}}}$, nonconformity scores E_i are computed:

$$E_i = \max\left(\hat{q}_{\alpha/2}(X_i) - Y_i, Y_i - \hat{q}_{1-\alpha/2}(X_i)\right)$$

The $(1 - \alpha)(1 + 1/n_{\text{cal}})$ -th empirical quantile of the scores, denoted as $Q_{1-\alpha}(E)$, is calculated. The final calibrated interval for a new prediction X_{n+1} is guaranteed to satisfy:

$$P\left(Y_{n+1} \in \left[\hat{q}_{\alpha/2}(X_{n+1}) - Q_{1-\alpha}(E), \hat{q}_{1-\alpha/2}(X_{n+1}) + Q_{1-\alpha}(E)\right]\right) \geq 1 - \alpha$$

7. Data Requirements and Schema Specifications

7.1 Entity-Relationship Overview

The operational data model links eight core entities:

- TrainMaster:** Canonical train definitions (Train Number, Name, Type, Priority Rank).
- StationMaster:** Physical station nodes (Code, Name, Latitude, Longitude, Zone, Division, Platform Count).
- TrackSegment:** Directed edges connecting stations (From Station, To Station, Length, MPS, Track Count, Electrification).
- ScheduleEntry:** Static timetable records (Train ID, Station ID, STA, STD, Distance, Day Count).
- LiveTelemetry:** Streaming locomotive reports (Loco ID, Train ID, Latitude, Longitude, Speed, Heading, Timestamp).
- CautionOrder:** Active speed restrictions (Segment ID, Start KM, End KM, Speed Cap, Reason, Expiration).
- WeatherAlert:** Local environmental conditions (Station/Zone ID, Visibility Meters, Fog Level, Timestamp).
- PredictionLog:** Generated arrival forecasts (Train ID, Station ID, Realized Time, Point ETA,

Lower Bound, Upper Bound, Generated Timestamp).

7.2 Data Schemas

7.2.1 Ingest Telemetry Payload (JSON)

JSON

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "RTISTelemetryPacket",
  "type": "object",
  "properties": {
    "packet_id": { "type": "string", "format": "uuid" },
    "loco_id": { "type": "string", "pattern": "^[0-9]{5}$" },
    "train_number": { "type": "string", "pattern": "^[0-9]{5}$" },
    "timestamp_utc": { "type": "string", "format": "date-time" },
    "coordinates": {
      "type": "object",
      "properties": {
        "latitude": { "type": "number", "minimum": 6.0, "maximum": 38.0 },
        "longitude": { "type": "number", "minimum": 68.0, "maximum": 98.0 },
        "altitude_m": { "type": "number" },
        "fix_quality": { "type": "string", "enum": ["GAGAN_DIFFERENTIAL", "GPS_3D", "CELLULAR_ESTIMATE"] },
        "required": ["latitude", "longitude", "fix_quality"]
      }
    },
    "kinematics": {
      "type": "object",
      "properties": {
        "speed_kmh": { "type": "number", "minimum": 0.0, "maximum": 180.0 },
        "heading_degrees": { "type": "number", "minimum": 0.0, "maximum": 360.0 },
        "required": ["speed_kmh", "heading_degrees"]
      }
    },
    "event_type": {
      "type": "string",
      "enum": ["PERIODIC_30S", "GEOFENCE_STATION_ARRIVAL", "GEOFENCE_STATION_DEPARTURE", "UNSCHEDULED_HALT"]
    }
  }
}
```

```
"required" "packet_id" "loco_id" "train_number" "timestamp_utc" "coordinates" "kinematics"
"event_type"
|
```

7.2.2 Prediction API Response Payload (JSON)

```
JSON
|
"$schema" "https://json-schema.org/draft/2020-12/schema"
"title" "DynamicETAPredictionResponse"
"type" "object"
"properties"
  "train_number" "type" "string" "example" "12951"
  "train_name" "type" "string" "example" "MUMBAI RAJDHANI"
  "current_status"
  "last_reported_station" "type" "string" "example" "BVI"
  "current_delay_minutes" "type" "integer" "example" 14
  "as_of_timestamp" "type" "string" "format" "date-time"
  |
  "predictions"
  "type" "array"
  "items"
  "type" "object"
  "properties"
    "station_code" "type" "string" "example" "MMCT"
    "scheduled_arrival_time" "type" "string" "format" "date-time"
    "predicted_eta_point" "type" "string" "format" "date-time"
    "predicted_delay_minutes" "type" "integer" "example" 8
    "confidence_interval_80pct"
    "type" "object"
    "properties"
      "lower_bound_eta" "type" "string" "format" "date-time"
      "upper_bound_eta" "type" "string" "format" "date-time"
      "margin_minutes" "type" "integer" "example" 4
    |
  "required" "lower_bound_eta" "upper_bound_eta" "margin_minutes"
  |
  "recovery_factors_applied"
  "type" "object"
```

```

    "properties"
    "engineering_allowance_min" "type" "number" "example" 3.5
    "traffic_recovery_time_min" "type" "number" "example" 5.0
    "caution_order_penalty_min" "type" "number" "example" 2.5
    }
    }
    }
    "required" "station_code" "scheduled_arrival_time" "predicted_eta_point"
    "confidence_interval_80pct"
    }
    }
    }
    "required" "train_number" "current_status" "predictions"
    }

```

8. Verification, Acceptance Criteria, and Traceability

8.1 Requirements Traceability Matrix (RTM)

Requirement ID	Module / Feature Name	Validation Methodology	Target Verification Criteria
FR-ING-01	High-Frequency RTIS Telemetry Ingestion	Stress and Load Testing	Sustained ingestion of 10,000 parallel streams at \$< 500\$ ms latency.
FR-FEA-01	Geospatial Track Map Matching	Automated Integration Testing	\$\ge 99.5\%\$ correct track alignment matching across benchmark routes.
FR-MOD-01	Macro RSTGCN Graph Embeddings	Performance Profiling	Network graph embedding refreshed every 60 seconds within 4 GB GPU VRAM.
FR-MOD-02	Micro Quantile Running Time	Regression Testing	Batch inference completed within

	Inference		50 ms per train query.
FR-MOD-04	Conformal Prediction Coverage Calibration	Statistical Validation	Empirical test coverage $\geq 80\%$ under Mondrian stratification.
FR-RUL-01	Dynamic Engineering Allowance Adjustment	Unit / Rule Engine Testing	Accurate recovery deduction when TSR is inactive; speed penalty applied when active.
FR-RUL-04	Fog-PASS Speed Cap Enforcement	Scenario Simulation	Maximum sectional velocity automatically capped at 60–75 km/h in fog zones.
FR-API-02	Passenger Public Enquiry REST API	API Performance Testing	P99 latency ≤ 200 ms under 20,000 requests/second load profile.
NFR-REL-01	Automated Fallback and Failover	Chaos Engineering / Fault Injection	Immediate automated fallback to baseline models upon artificial ML serving failure.

8.2 Operational Acceptance Benchmarks

The dynamic prediction system achieves production operational acceptance upon satisfying the following performance criteria across a 30-day trial period on a designated trunk corridor (e.g., Delhi–Kanpur–Prayagraj–Howrah):

1. **Prediction Error Reduction:** Demonstrates an overall Mean Absolute Error (MAE) reduction of $\geq 15\%$ compared to the legacy static schedule extrapolation used by the baseline enquiry system.
2. **Interval Coverage Validity:** Maintains a Prediction Interval Coverage Probability (PICP) $\geq 80\%$ for conformalized bounds across all evaluated coaching services.

3. **Punctuality Impact:** Generates advance platform throat conflict alerts that support a measurable reduction in terminal yard approach holding delays.
4. **Availability Compliance:** Maintains $\geq 99.99\%$ continuous service uptime without data loss or pipeline interruption during scheduled failover operations.